

4.1. The formula of the first homology groups. We recall the construction of abelian branched coverings of (M, L) . Let G be a finite abelian group and $\pi : \pi_1(M - L) \rightarrow G$ be a surjective homomorphism. Let $M_\pi \rightarrow M$ denote the branched cover corresponding to π , that is, the Fox completion of the unbranched covering corresponding to $\ker \pi$. Then the transformation group of $M_\pi \rightarrow M$ is G . Note that any finite abelian cover of M branched along L is obtained in this way.

We also recall the formula for the size of the first homology groups of abelian branched coverings given by Mayberry–Murasugi and Porti. Let $\hat{G}$ denote the set of homomorphisms from G to $\mathbb{C}^*$. We choose meridians $m_1, \dots, m_d \in \pi_1(M - L; \mathbb{Z})$. For $\xi \in \hat{G}$, we define a sublink L_ξ of L as

$$L_\xi = \cup_{\xi(\pi(m_i)) \neq 1} l_i.$$

We write $L_\xi = l_{i_1} \cup \dots \cup l_{i_k}$. For the trivial homomorphism $1 \in \hat{G}$, we define $L_1 = \emptyset$ and $\Delta_{L_1} = 1$. We also set

$$\hat{G}^{(1)} = \{\xi \in \hat{G} \mid L_\xi \text{ has a single component}\}.$$

We write $i(\xi)$ for the index of the meridian corresponding to $\xi \in \hat{G}^{(1)}$. For a group H , let $|H|$ be the size of H if H is finite and 0 if H is infinite. Then Mayberry–Murasugi and Porti gave the following theorem.

Theorem 4.1 ([8, Theorem 10.1], [4, Theorem 1.1]).

$$|H_1(M_\pi; \mathbb{Z})| = \frac{|G|}{\prod_{\xi \in \hat{G}^{(1)}} |1 - \xi(\pi(m_{i(\xi)}))|} \prod_{\xi \in \hat{G}^{(1)}} |\Delta_{L_\xi}(\xi(\pi(m_{i_1})), \dots, \xi(\pi(m_{i_k})))|. \quad (4.1)$$

4.2. $\mathbb{Z}_p^d$ -coverings. Let $\pi : \pi_1(M - L) \rightarrow \mathbb{Z}_p^d$ be the composition map of the abelianization $\pi_1(M - L) \rightarrow \pi_1(M - L)^{ab} (\cong \mathbb{Z}^d)$ and the inclusion $\mathbb{Z}^d \hookrightarrow \mathbb{Z}_p^d$. For a finite index subgroup $\Gamma \subset \mathbb{Z}_p^d$ let $\pi_\Gamma : \pi_1(M - L) \rightarrow \mathbb{Z}_p^d/\Gamma$ denote the composition of π and the canonical surjective map $\mathbb{Z}_p^d \rightarrow \mathbb{Z}_p^d/\Gamma$. Let $M_\Gamma \rightarrow M$ denote the branched covering of (M, L) corresponding to π_Γ .

Definition 4.2. A branched $\mathbb{Z}_p^d$ -covering of (M, L) is the inverse system of branched coverings $M_\Gamma \rightarrow M$ where Γ runs finite index open subgroups in $\mathbb{Z}_p^d$.

We obtain the p -adic convergence of the non- p -parts of the sizes of the first homology groups in branched $\mathbb{Z}_p^d$ -covering of (M, L) . For a $\mathbb{Z}_p$ -module A let $\text{rank } A = \dim_{\mathbb{F}_p} A \otimes \mathbb{F}_p$.

Lemma 4.3. Let $\Gamma_1 \supset \Gamma_2 \supset \dots$ be a descending sequence of finite open subgroups such that $\cap_n \Gamma_n = \{0\}$ in $\mathbb{Z}_p^d$. Then we have

- (1) for all sufficiently large n , $\text{rank } \mathbb{Z}_p^d/\Gamma_n = d$.
- (2) for each meridian m of L , $\lim_{n \rightarrow \infty} \text{ord}(\pi_{\Gamma_n}(m)) = \infty$.

Proof. We put $G_n = \mathbb{Z}_p^d/\Gamma_n$.

(1) We note that $\varprojlim_n G_n \cong \mathbb{Z}_p^d$ by the assumption $\cap_n \Gamma_n = \{0\}$. Since the canonical homomorphisms $F_n : \mathbb{Z}_p^d \rightarrow G_n$ and $f_{n/n-1} : G_n \rightarrow G_{n-1}$ are surjective the induced homomorphisms $\tilde{F}_n : \mathbb{Z}_p^d \otimes \mathbb{F}_p \rightarrow G_n \otimes \mathbb{F}_p$ and $\tilde{f}_{n/n-1} : G_n \otimes \mathbb{F}_p \rightarrow G_{n-1} \otimes \mathbb{F}_p$ are also surjective. Thus we have that the sequence $(\text{rank } G_n)$ is monotonically increasing. We suppose that $\text{rank } G_n < d$ for all $n > 0$. Then we have that $\dim_{\mathbb{F}_p} \ker \tilde{F}_n \geq 1$ for all n . Since $\ker \tilde{F}_n \subset \ker \tilde{F}_{n-1}$ for all $n > 1$, we see that $\dim_{\mathbb{F}_p} \cap_n \ker \tilde{F}_n \geq 1$. Thus